
ME 611: Design and Optimization of Thermal Systems

Course Code	ME 611
Course Name	Design and Optimization of Thermal Systems
L-T-P-C	3-0-0-3
Prerequisites	Thermodynamics, Fluid Mechanics, Heat Transfer
Intended for	MS/PhD
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- Introduction to system design – Regression analysis and curve fitting – modeling of thermal equipment – system simulation (successive substitution – Newton – Raphson method) – examples – economic analysis – optimization – Lagrange multipliers, search methods, linear programming, geometric programming – New generation optimization techniques – simulated annealing, Genetic Algorithms, Bayesian statistics.
- Examples applied to heat transfer problems and energy systems such as gas and steam power plants, refrigeration systems, heat pumps and so on.

Text and Reference Books

1. Y.Jaluria, **Design and optimization of thermal systems**, McGraw Hill, 1998.
2. Adrian Bejan, George Tsatsaronis, Michael Moran, **Thermal Design and Optimization**, John Wiley and Sons, 1995.
3. L.C. Burmeister, **Elements of thermal fluid system design**, Prentice Hall, 1998.
4. W.F.Stoecker, **Design of thermal systems**, McGraw Hill, 1989.